BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 **Data Analysis Report**

Project ID: 25-26J-106

Project Title: Adaptive And Explainable Human-Robot Collaboration For Intelligent Sorting Systems

1. Introduction

1.1. Background

Human-robot collaboration (HRC) is an exciting area to explore in intelligent interactive systems. There is a need for adaptive and multi-sensory approaches in education for differently abled students apart from traditional approaches. The integration of robotics with computer vision and machine learning will create interactive environments where the student learns through observation and interaction. Accurate material recognition is difficult in robotic sorting because of various textures and visual appearances [1],[2]. The system employs a vision-based texture recognition algorithm using DINOv2 [3] for classifying the materials into plastic, metal, glass, and paper with confidence scores. Region-of-interest segmentation is employed, and the object is moved to improve visibility. Successful HRC should be flexible in adapting to the actions of the human and interacting in real-time [4],[5]. Here we present a robot-assisted learning system which is useful to differently abled children and normally developed children. It combines a vision-based pattern completion task with speech repetition [6]. Intuition plays a vital role as humans indicate their intentions using eye and head gestures [7]. The technology used includes a novel gaze and head pose-based object selection technique with the use of one RGB camera. Gaze and head pose information together with object detection enable the system to determine users' intentions, providing a solution without any specialized sensors, making the approach efficient and affordable [8]. Transparency and adaptability of the robotic behavior are critical to achieving positive effects when working with educational purposes [9]. The sorting policy used in the system is recommendation-driven; that is, actions are performed according to object properties detected in real-time. Moreover, explainability provided by the system in the form of XAI allows for understanding the robot's reasoning, which makes both teachers and pupils engaged [10]. As a result, it becomes possible for the latter to not only perform certain tasks but also analyze how they should be done. Driven by the lack of adaptive and interactive platforms for differently abled kids, the proposed solution integrates multiple techniques to support multi-modal learning.

1.2. Research Problem

Despite some progress in HRC, robot perception, gaze-based interaction, recommendation systems, and XAI, the problem of building a comprehensive robotic solution that can work in real-time with perception, interaction, intelligent decision making, and explainability capabilities in the context of education for disabled children remains unresolved.

Most of the current HRC systems are designed for industrial applications and cannot provide personalized support based on object recognition capabilities. Even though today's robot perception techniques are able to recognize different objects efficiently, they find it difficult to identify textures and materials necessary for educational purposes. Moreover, gaze-based interaction, recommendation, and XAI solutions are created independently from each other. Recommendations tend to be more task-oriented than educational, whereas available XAI methods generate explanations off-line rather than in real-time.

Therefore, there is a need for an integrated human-robot collaborative system that combines,

- Real-time object and material recognition using deep visual learning
- Multi-modal sensing and interaction
- Gaze-based human intention detection
- Intelligent recommendation-based decision making
- Real-time explainable AI for transparent educational assistance

Such a system would improve interactive learning experiences and provide effective support for differently abled children through adaptive and explainable robotic assistance.

1.3. Objectives

It Number	Objective	Objective number
IT22238962	To develop a deep learning–based system to classify rubber, plastic, metal, and glass, enhanced with sensor fusion and deployed in a real-time robotic platform for assistive applications.	1

IT22089618	Develop real-time object detection. Build speech recognition and classification system. Integrate robotic arm for physical learning interaction. Develop multimodal adaptive learning system	2
IT22078032	To implement a gaze-guided selection mechanism enabling natural, hands-free human–robot interaction	3
IT22888716	Design a YOLOv8 object detection algorithm, develop a machine learning recommendation algorithm for sorting purposes, deploy Explainable AI (SHAP), and automate robotic sorting with a robotic arm controlled by Arduino programming language.	4

2. Data Exploration

2.1. Data Collection

1) Collaborative Texture Recognition

Data was collected in a controlled laboratory environment using an RGB camera implemented via the IP Webcam application on a smartphone. This low-cost setup allowed flexible camera positioning across different angles, distances, and lighting conditions. Objects representing four material categories — plastic, glass, rubber, and metal — as well as an "other" category for unclassified items, were photographed.
Images were captured from multiple viewpoints: top shots, side shots, and angled views to ensure the neural network learned material properties rather than object geometry alone. Variations in size, shape, and surface quality (shiny, matte, smooth, rough) were deliberately included to improve generalization.

2) Adaptive Pattern Recognition via Human Behavior

Data for this research was collected through a combination of custom dataset creation and real-world observations. A set of images containing colored objects such as balls, boxes, and balloons was captured under varying environmental conditions including different lighting, angles, and backgrounds. These images were used to train the object detection model. Additionally, speech data was collected in the form of recorded verbal responses from users, which were later transcribed using the Whisper speech recognition system. Stakeholder input from SERD Kelaniya Special Needs School also contributed to defining realistic data requirements and system behavior.

3) Human-Led Selection Through Gaze-Guided Robotic Assistance

Data was collected using standard RGB cameras and CCTV cameras positioned in a fixed frontal orientation relative to the user. The cameras captured facial images and video frames of participants looking at predefined regions within the workspace.

The data collection process involved,

- Placing objects in fixed spatial regions (left, right, center, up, down, etc.)
- Instructing participants to look at specific regions for controlled durations
- Capturing facial frames continuously using CCTV/RGB cameras
- Extracting frames from video streams at fixed intervals
- Labeling each frame according to the corresponding gaze direction

No specialized hardware such as eye trackers or depth sensors was used.

4) Integrating Explainable AI and Adaptive Recommendations for Human Centered Robotic Sorting

The data collection was done through a controlled setting with the aid of an RGB camera that was created by means of IP Webcam application in the mobile phone. Through this cheap method, images could be taken with varying positions and distance as well as varied lighting and background conditions for the objects in order to create different settings. The objects used to create the images belong to different categories according to their shapes and colors, like ball, box, pyramid, red, green, and blue.

The position of the camera when taking images was not just the usual top-down view but varied between top, side, and oblique positions to ensure the creation of generalized features of the objects by the model regardless of position.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
RGB Image Dataset	Images of objects across 5 material categories: plastic, glass, rubber, metal, and "other"	Captured using IP Webcam (smartphone RGB camera) in a controlled lab environment	Total Images: 8351 Number of Classes: 5	Material label, surface texture, reflectance, transparency, shape variation, lighting conditions
Ultrasonic Sensor Data	Distance and object presence readings collected during robotic operation	HC-SR04 ultrasonic sensor connected to microcontroller	Real-time streaming data	Object presence (Boolean), distance measurement (cm), detection reliability, false-positive rate
Classroom Image Data	Image data collected from classroom interactions to detect objects and analyze student behavior patterns.	Camera-based data collection	Medium (multiple labeled images)	Object type, object position, bounding boxes, visual patterns
Speech Data	Audio data collected from students to analyze verbal responses and interaction.	Microphone recordings	Medium (audio samples)	Speech content, transcribed text, keywords

Custom Labeled Dataset	A manually created dataset of learning objects used for training object detection and pattern recognition models.	Manually annotated dataset	Medium	Object classes, labels, pattern sequences
RGB camera and CCTV camera	Self-collected facial images and video frames capturing eye gaze direction and head pose across predefined spatial regions for gaze-based interaction analysis.	Primary data	1081	Eye gaze direction, head pose orientation, face bounding box
Custom Object Image Dataset for Intelligent Sorting	A self-collected image dataset created to support real-time object detection and recommendation generation for intelligent sorting. Images consist of physical objects arranged in a controlled environment, capturing variations in object color and shape relevant to	Captured using ESP32 Camera in a controlled experimental setup. Image annotations were generated using Roboflow.	2080	Object Class (Color + Shape), Bounding Box Coordinates (x-center, y-center, width, height), Object Count per Frame

Human-Guided Object Sorting Recommendation Dataset	industrial sorting scenarios. Each image is paired with an automatically generated annotation file in YOLO format, describing object class and spatial location. Human-Guided Object Sorting Recommendation Dataset	Manually generated by the researcher using detected object counts derived from ESP32 camera images and YOLO outputs	9000	Image Identifier, Object Count per Color (Blue, Green, Red), Object Count per Shape (Box, Ball, Pyramid), Human-Selected Sorting Strategy

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

	1	2	3	4
combined RGB and CCTV image dataset	The collected datasets are highly suitable for achieving the objectives of the proposed collaborative texture recognition system. The RGB image dataset provides diverse visual information related to texture,	The datasets used in this research are highly relevant to the defined objectives. The image dataset directly supports the development of the object	The combined RGB and CCTV image dataset, consisting of 1,081 self-collected facial images, directly supports the research objective of developing a non-intrusive, vision-based gaze estimation system using standard	The study presents an intelligent robotic sorting system that combines computer vision, machine learning, and Explainable AI (XAI) to support human-robot

	reflectance, transparency, shape, and lighting conditions, which are essential for training deep learning models to accurately classify materials such as plastic, glass, rubber, and metal. The inclusion of multiple viewpoints and environmental variations improves the robustness and generalization capability of the models in real-world scenarios. Furthermore, the dataset supports the comparison of different architectures including CNN-based models and transformer-based models such as DINOv2 for real-time material recognition tasks. In addition, the ultrasonic sensor dataset enhances the system by providing reliable object presence and distance measurements, which contribute to multimodal sensor	detection model and enables accurate identification of objects within the learning environment. It also contributes to robotic interaction by providing positional information required for pattern demonstration. The speech dataset is essential for evaluating verbal responses and supports the development of the classification model. Both datasets collectively contribute to achieving the multimodal integration objective, as they enable seamless interaction between visual, verbal, and physical	cameras. Since both RGB and CCTV images were collected under realistic conditions, the dataset effectively represents the visual constraints present in real production and surveillance environments, aligning closely with the core research problem. The labeled images enable supervised learning for eye gaze and head pose classification, fulfilling the objective of accurately estimating gaze direction without specialized hardware. In addition, the video data captured using RGB and CCTV cameras is used exclusively for region-based testing and validation, allowing evaluation of gaze stability, dwell time, and region selection consistency over time. This supports the objective of mapping gaze direction to predefined	collaboration in education, especially for differently-abled children. The system uses YOLOv8 for object detection and a Random Forest model for decision-making, achieving high accuracy (~94%) in selecting optimal sorting strategies. A key strength is the integration of explainability (via SHAP), allowing the robot to justify its decisions, which improves trust, transparency, and learning value for educators and students. The system also includes real-time robotic execution using Arduino, making it practical and interactive. Overall, the research

	fusion and improve the overall safety, accuracy, and responsiveness of the robotic platform. Together, these datasets effectively support real-time perception, intelligent robotic interaction, and assistive educational applications for differently abled students.	components of the system.	workspace regions for gaze-guided robot object selection. Overall, the dataset design ensures strong alignment between data collection, model training, and real-world deployment objectives.	effectively addresses gaps in real-time explainable robotics and educational applications, showing that combining AI + robotics + XAI can enhance both automation efficiency and learning experiences.

3. Methodology

Collaborative Texture Recognition

3.1.1 Data Preprocessing

Data preprocessing was performed to ensure consistency, improve model performance, and enhance generalization across varying conditions. All RGB images were resized to a standard input resolution and normalized by scaling pixel values to a range of 0 to 1. Data augmentation techniques, including rotation, horizontal flipping, brightness adjustment, and zooming, were applied to simulate real-world variability such as lighting changes and object orientations. Noise reduction was applied where necessary, and images were converted into tensor format for compatibility with deep learning models. For sensor data, ultrasonic readings were processed using thresholding to convert distance values into binary object presence indicators, while outliers and false positives were filtered to improve reliability.

3.1.2 Scalability

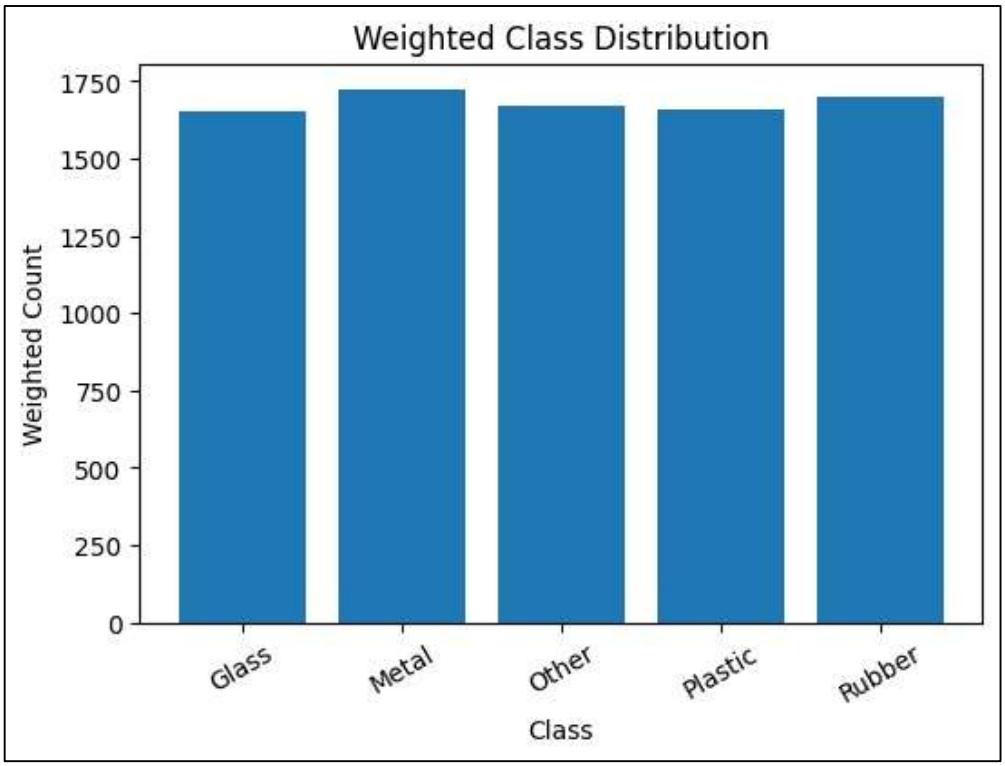
The proposed system is designed with scalability in mind, both in terms of data expansion and system architecture. Although the current dataset is sufficient for a proof-of-concept, it can be extended to include more material categories and diverse environmental conditions without significant changes to the pipeline. The use of a pre-trained self-supervised model such as DINOv2 enhances scalability by enabling strong generalization even with limited labeled data. Additionally, the modular structure of the system, comprising vision, sensor, and robotic control components allows each module to be independently upgraded or expanded. The lightweight nature of the ultrasonic sensor also supports integration into larger multi-sensor or multi-robot systems.

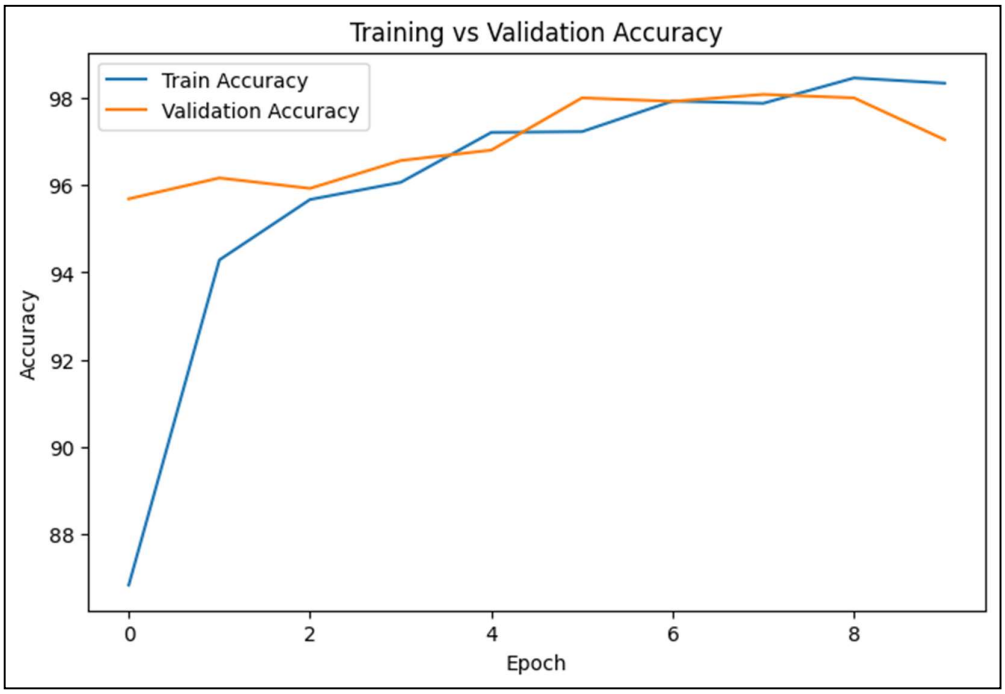
3.1.3 Feature Extraction

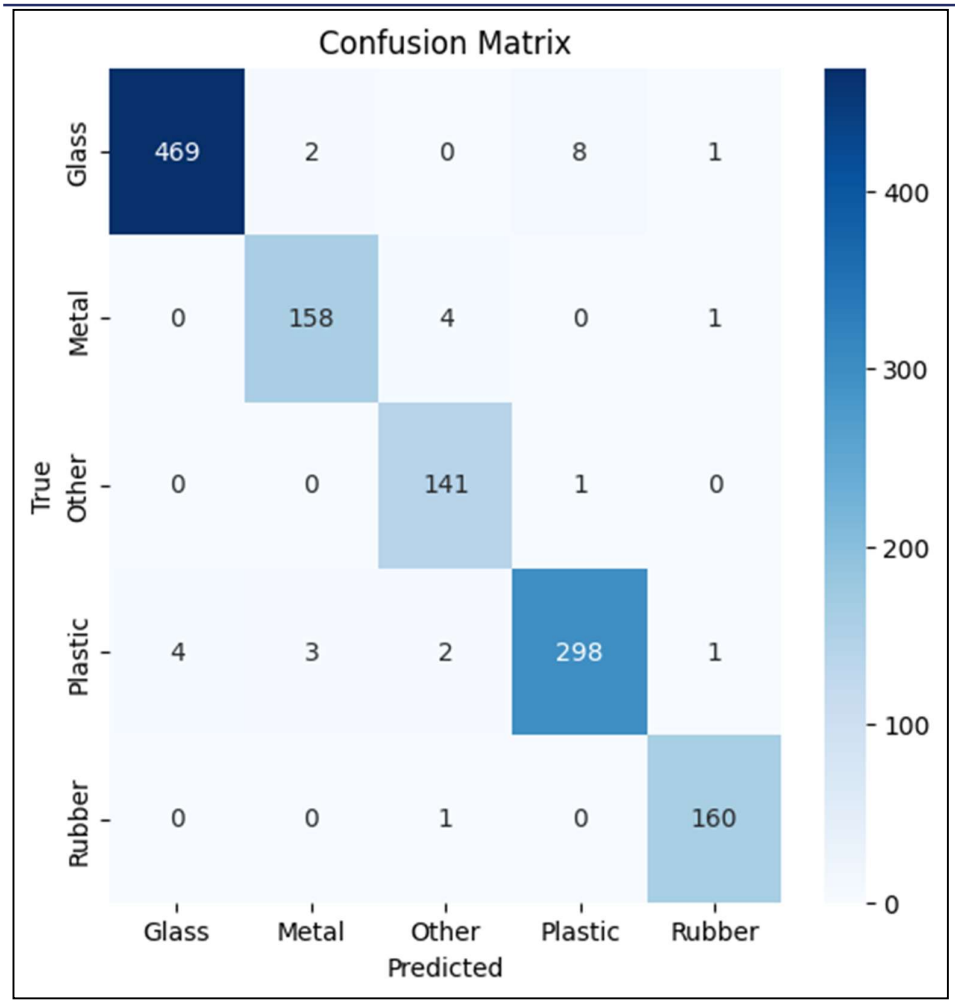
Feature extraction was carried out to capture meaningful patterns from both visual and sensor data for accurate material classification and system reliability. The DINOv2 vision transformer extracted both global and local features using attention mechanisms, enabling the model to identify key characteristics such as texture gradients, reflectance, and surface patterns. This approach outperformed traditional CNN-based models, which primarily rely on local feature hierarchies and often struggle with fine-grained distinctions. For sensor data, ultrasonic readings were used to extract object presence information, which acted as a gating signal to trigger classification. This combination of visual and sensor-based features improved overall system accuracy and robustness.

3.1.4 Key Insights

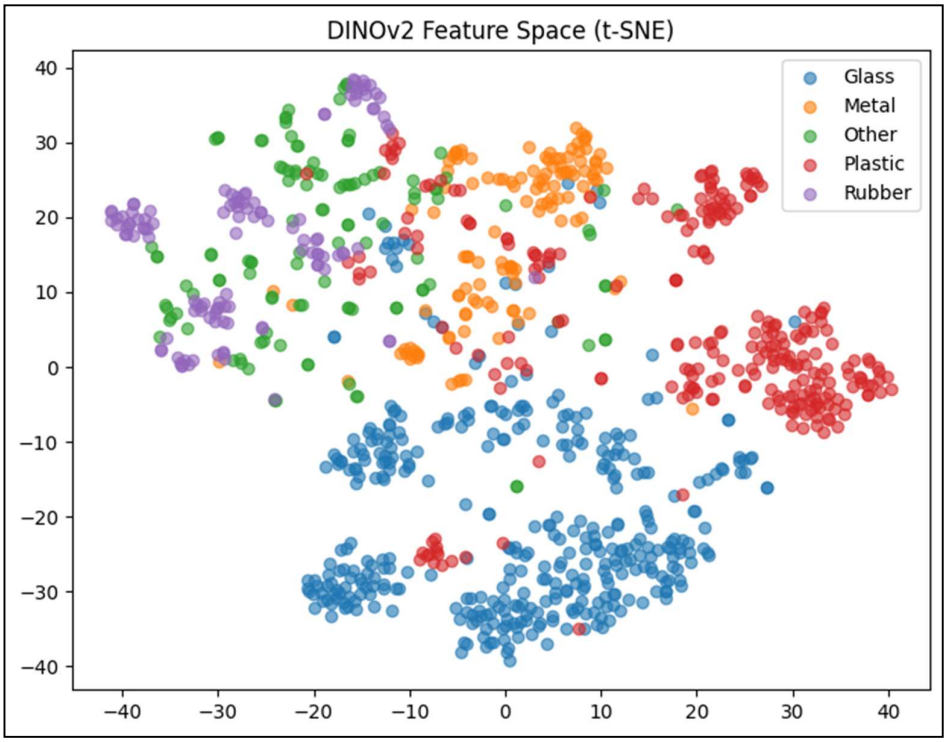
The analysis revealed that transformer-based models significantly outperform traditional CNN architectures in material texture recognition tasks. DINOv2 achieved the highest accuracy due to its ability to capture both global context and fine-grained details through self-attention mechanisms. CNN models such as Sequential CNN, EfficientNet, and MobileNet showed reasonable performance but struggled with visually similar materials and environmental variability. The integration of the ultrasonic sensor improved system reliability by ensuring classification is only triggered when an object is present, reducing false detections. Overall, the combination of advanced vision models and sensor fusion enabled a robust, real-time system suitable for assistive educational applications.


Weighted Class Distribution
Weighted Count
1750
1500
1250
1000
750
500
250
0
Glass
Metal
Other
Plastic
Rubber
Class


Training vs Validation Accuracy
Train Accuracy
Validation Accuracy
Accuracy
98
96
94
92
90
88
0
2
4
6
8
Epoch



	precision	recall	f1-score	support
Glass	0.99	0.98	0.98	480
Metal	0.97	0.97	0.97	163
Other	0.95	0.99	0.97	142
Plastic	0.97	0.97	0.97	308
Rubber	0.98	0.99	0.99	161
accuracy			0.98	1254
macro avg	0.97	0.98	0.98	1254
weighted avg	0.98	0.98	0.98	1254



Adaptive Pattern Recognition via Human Behavior

3.2.1. Data Preprocessing

Data preprocessing was carried out separately for image and speech datasets. For the image dataset, preprocessing included resizing images to match YOLO model requirements, normalizing pixel values, and converting annotations into YOLO format. Data augmentation techniques such as random flipping, scaling, and color adjustments were applied to improve model generalization. The dataset was split into training, validation, and testing sets using a 70:15:15 ratio.

For the speech dataset, preprocessing involved noise reduction, audio normalization, and speech-to-text conversion using the Whisper model. The resulting text was tokenized into structured object sequences. Feature engineering techniques were applied to extract meaningful features such as token overlap, sequence alignment, and response length differences, which were used for classification.

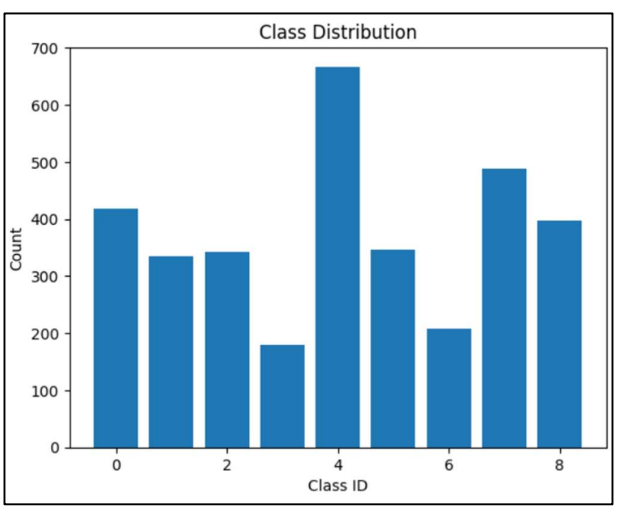
In addition, data integration techniques were applied to combine outputs from vision, speech, and robotic modules, ensuring synchronized operation within the multimodal system.

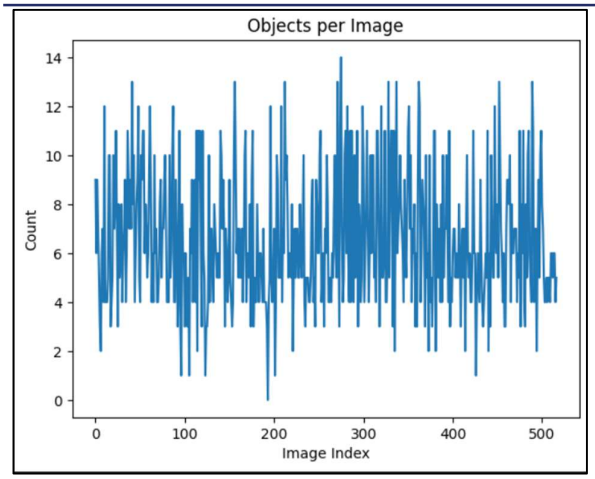
3.2.2. Scalability

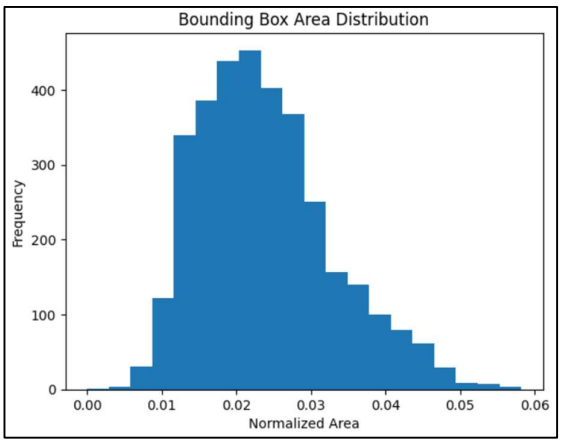
The datasets used in this research are scalable and can be expanded to improve system performance. The image dataset can be increased by capturing additional object variations and applying further augmentation techniques. Similarly, the speech dataset can be expanded by collecting more diverse verbal responses from users. The use of transfer learning in the YOLO model reduces the need for large datasets, making the system efficient even with moderate data sizes. The modular architecture of the system also supports scalability by allowing independent updates to each component.

3.2.3. Feature extraction

Feature extraction was performed to identify key attributes relevant to each objective. In the object detection module, features such as object class labels, bounding box coordinates, and spatial relationships were extracted. These features enable accurate detection and positioning of objects in real time. In the speech classification module, features such as token sequences, token overlap ratios, sequence order accuracy, and response length differences were extracted. These features allow the system to evaluate the correctness of verbal responses effectively.







Human-Led Selection Through Gaze-Guided Robotic Assistance

3.3.1. Data Preprocessing

The combined RGB and CCTV image dataset underwent several preprocessing steps to ensure quality and consistency. Data cleaning was performed by removing blurred, low-quality, and partially occluded images, as well as frames where faces were not clearly detectable. The RGB and CCTV images were then integrated into a unified dataset with a consistent labeling structure. Gaze directions were mapped to predefined spatial regions aligned with the robotic workspace, and labels were encoded into numerical classes for supervised learning. All images were resized to a fixed resolution, converted to floating-point format, and normalized to ensure stable model training. Missing or invalid samples, such as those with failed face detection or missing labels, were removed rather than imputed, maintaining dataset integrity.

3.3.2. Scalability

The combined RGB and CCTV image dataset, consisting of 1,081 self-collected facial images, is sufficient for developing and validating a research-level prototype for gaze direction and head pose classification. The dataset provides balanced representation across predefined gaze regions, enabling effective supervised learning and model evaluation.
The data collection methodology is inherently scalable, as it relies solely on standard RGB and CCTV cameras that are commonly available in real-world environments. The dataset can be expanded by:

- Recording additional participants
- Increasing capture duration
- Extracting more frames from video streams
- Integrating multiple camera viewpoints

Furthermore, the use of automated preprocessing and CNN-based feature extraction allows the system to handle larger datasets without significant modification. This ensures the approach is scalable for extended experimentation and potential real-world deployment in production environments.

3.3.3. Feature Extraction

Feature extraction was performed using a deep learning-based approach, where discriminative features were automatically learned from facial images by the convolutional neural network.

Feature Extraction Steps

1. Face Region Isolation
    - Facial regions were detected and cropped from raw images
    - Ensured focus on eye and head pose information
2. Input Standardization
    - Cropped facial images were resized to a fixed resolution
    - Pixel values were normalized prior to model input

3. Deep Feature Learning
    - A ResNet-18 convolutional neural network was used to extract hierarchical visual features
    - Lower layers captured edge and texture information
    - Deeper layers captured gaze direction and head orientation cues
4. Feature Evaluation
    - Extracted features were evaluated indirectly through classification performance

- Model accuracy and confusion matrices were used to assess feature discriminability across gaze regions

5. Region-Based Feature Mapping
   - Predicted gaze classes were mapped to predefined workspace regions
   - Supported gaze-guided region selection for robotic object picking

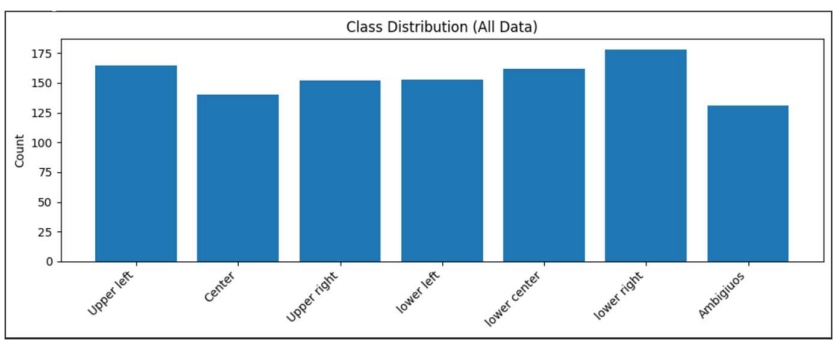
   This automated feature extraction approach eliminates the need for manual feature engineering and enables robust gaze estimation using only RGB visual data.

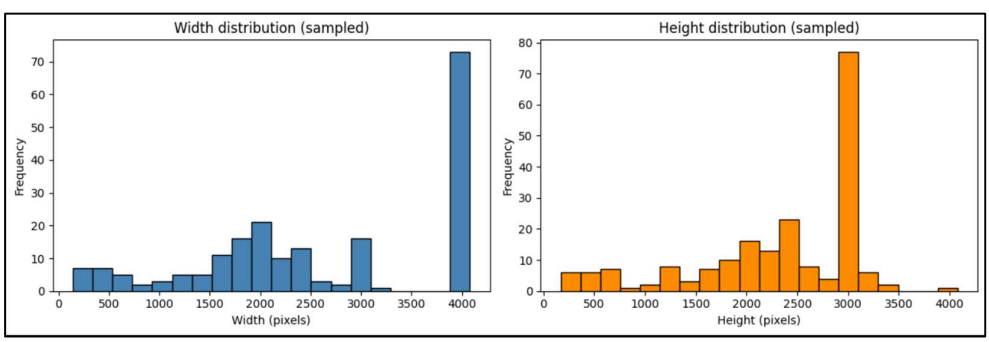
3.3.4. Key Insights

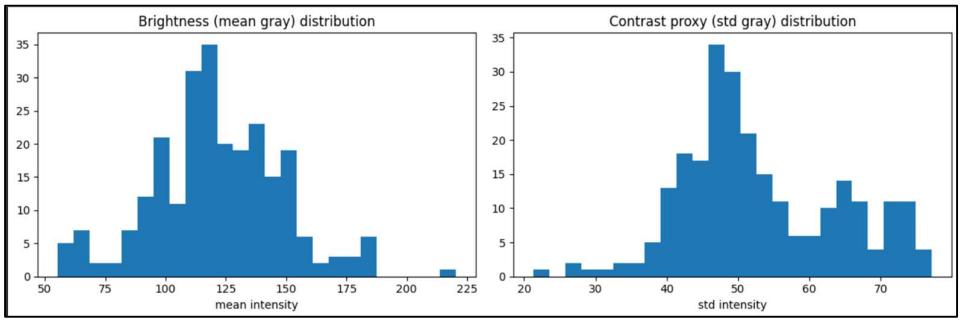
Dataset Exploration and Distribution

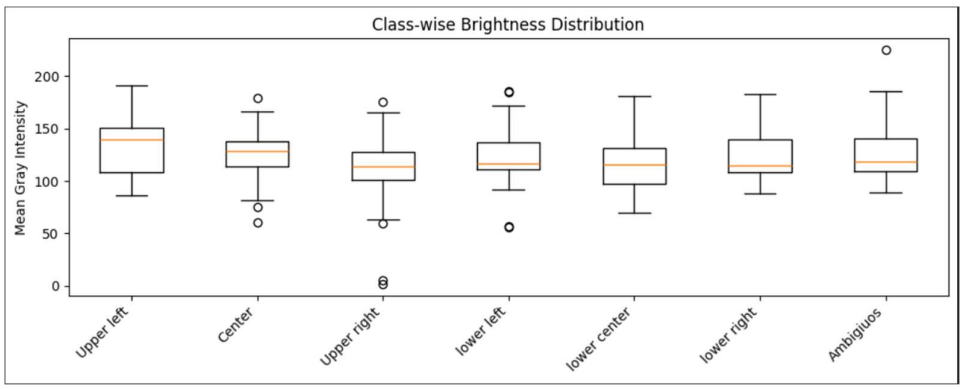
Exploratory data analysis revealed that the combined RGB and CCTV image dataset (1,081 images) is balanced across predefined gaze regions, ensuring unbiased model training. Class distribution visualizations confirmed approximately uniform representation across all gaze directions, reducing the risk of class dominance.

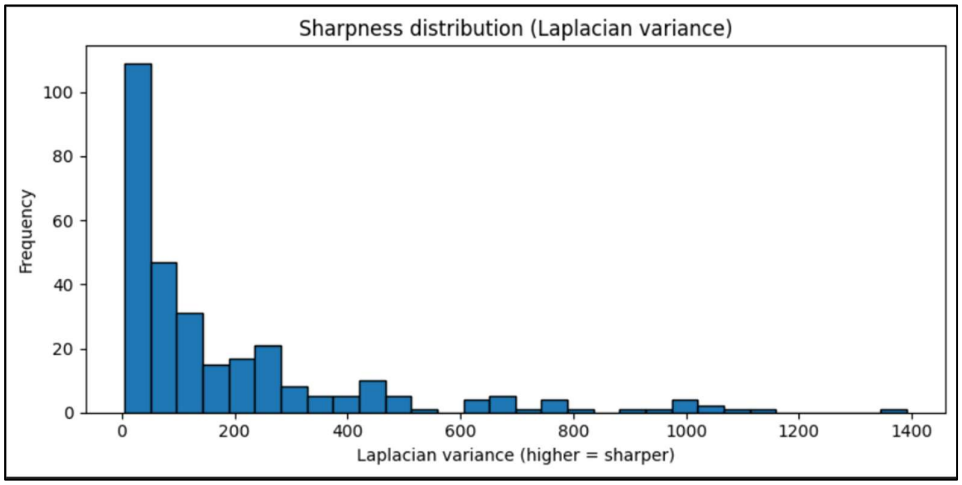
Sample image visualizations highlighted noticeable differences between RGB and CCTV data in terms of resolution and lighting. Despite these variations, facial features relevant to eye gaze and head pose remained sufficiently visible for learning discriminative patterns.

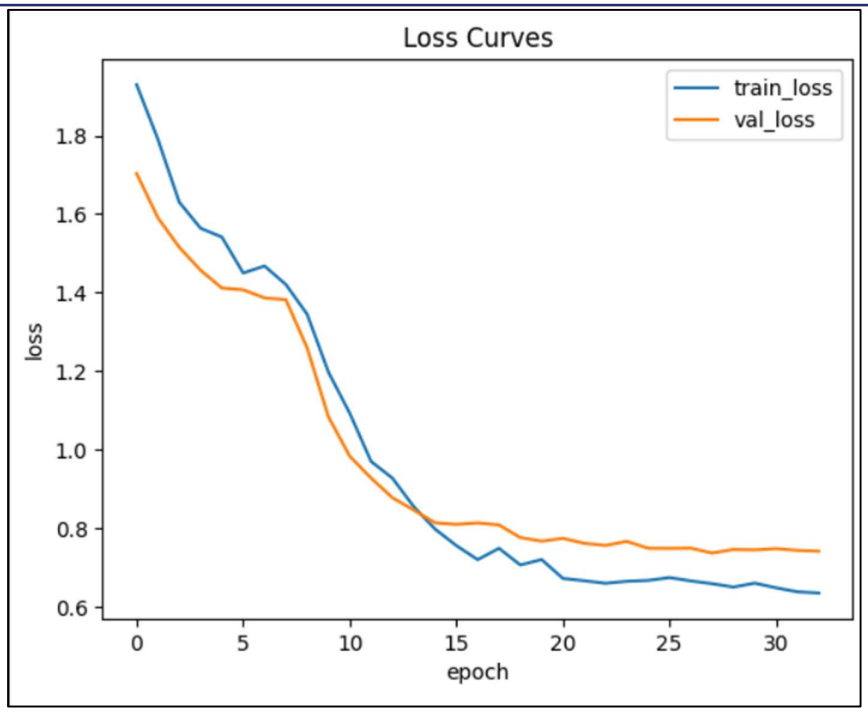


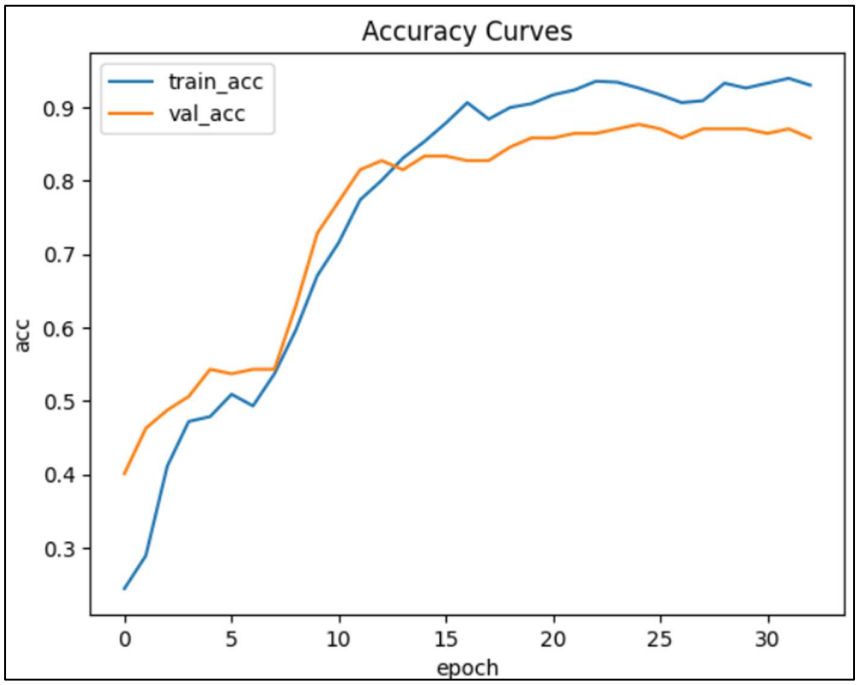



Brightness (mean gray) distribution
mean intensity
Contrast proxy (std gray) distribution
std intensity


Class-wise Brightness Distribution
Mean Gray Intensity
Upper left
Center
Upper right
lower left
lower center
lower right
Ambigiuos


Sharpness distribution (Laplacian variance)
Frequency
Laplacian variance (higher = sharper)


Loss Curves
train_loss
val_loss
loss
epoch
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0
5
10
15
20
25
30


Accuracy Curves
train_acc
val_acc
acc
epoch
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
5
10
15
20
25
30

Integrating Explainable AI and Adaptive Recommendations for Human Centered Robotic Sorting

3.4.1 Image Dataset preparation

A dataset of 2080 images was created to train the object detection model. Multiple objects in each image are placed in ways that simulate sorting. These objects incorporate attributes such as color (blue, green, and red) and shape (box, ball, and pyramid), which may occur during the sorting of real-world objects.

Each image was annotated using YOLO annotation format. The dataset was split into training and validation sets based on a specific configuration to have organized and reproducible training.

3.4.2 Train the Dataset

The YOLOv8 model was chosen as the object detection model because of its excellence in real-time object detection tasks. To optimally combine detection precision and efficiency, the lightweight YOLOv8n model was selected for this purpose.

The model was trained with pre-trained weights that were further fine-tuned on the customized dataset for 50 epochs with a batch size of 16. This trained model is subsequently used as the perception module while carrying out real-time processing. The trained model has been capable of learning object concepts based on color and shape.

3.4.3 Extraction of Features at Scene Levels

Instead, it does not use individual bounding boxes for decision-making directly but instead aggregates detection results at the scene level. YOLOv8 summarizes each input frame it processes by counting the detected objects. Each scene is represented using six interpretable features:

- count_blue
- count_green
- count_red
- count_box
- count_ball
- count_pyramid

This representation captures object dominance within the scene, thereby being extremely close to human reasoning. This feature formulation based on dominance allows an intuitive interpretation and gives good support for explainability since the features obviously show which object attributes are driving the sorting of objects.

3.4.4 Strategy Recommendation Dataset Construction

A dataset of strategy recommendations consisting of 2,520 labeled scenes was constructed. Each scene contains the six extracted features, along with an associated strategy label representing the preferred first sort method.

The strategy labels are derived from object dominance, including color-based and shape-based strategies like blue_first and ball_first. These are designed to reflect real human decision-making behavior, which tends to rely more on perceived importance rather than on strict rules. Because of this, the dataset carries realistic preferences that could be observed in human sorting.

3.4.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed in order to understand the data structure, distributions, and relationships in the dataset regarding strategy recommendations. It helped in understanding feature trends, class distribution, and inter-variable relationships.

3.4. 6 Scalability

The dataset size (2,080 images) is sufficient for training the YOLOv8 model with transfer learning. The structured dataset used for machine learning is scalable and can be expanded by generating additional scenarios. The system can be scaled further by adding more object types and increasing dataset diversity.

3.4.7 Feature Distribution Across Strategies

Box plot analysis was conducted to see how the object counts were distributed across strategies. It was found that every strategy has its distinct form of object count distribution. For instance, scenes marked as blue_first usually had larger object counts of blue objects.

However, there was also a significant amount of overlap between strategies, especially in cases where more than one object fought for dominance in an image. The amount of overlap between strategies points to the limitations of rule-based systems, where simple decision-making rules may not always guarantee the correct identification of strategies. The paper underlined the need for machine learning approaches capable of processing overlaps

3.4.8 Strategy-Wise Feature Averages

Feature averages for each strategy were compared and visualized in a heatmap. This analysis found that the naming of strategies is consistent with the dominant features observed in the data. Consecutively, this consistency supports internal validity of the dataset and promotes its suitability for predictive modeling.

3.4.9 Multivariate Analysis

Pairwise feature plots and PCA were used to evaluate class separability in a reduced-dimensional space. While partial clustering of strategies was observed, significant overlap remained. This reflects inherent ambiguities in human decisionmaking where no single object category clearly dominates a scene. The analysis reinforced the complexity of the strategy recommendation task.

3.4.10 Data Preparation and Balancing

The data was split into training and testing samples using stratified sampling, which helped in ensuring that the samples were well represented in their actual strategy distribution.

To overcome the issue of imbalance in the training dataset, the Synthetic Minority Oversampling Technique (SMOTE) was only used for the training dataset. The test dataset was not altered. This approach enabled the model to be trained well by SMOTE but avoided biased results for the assessment.

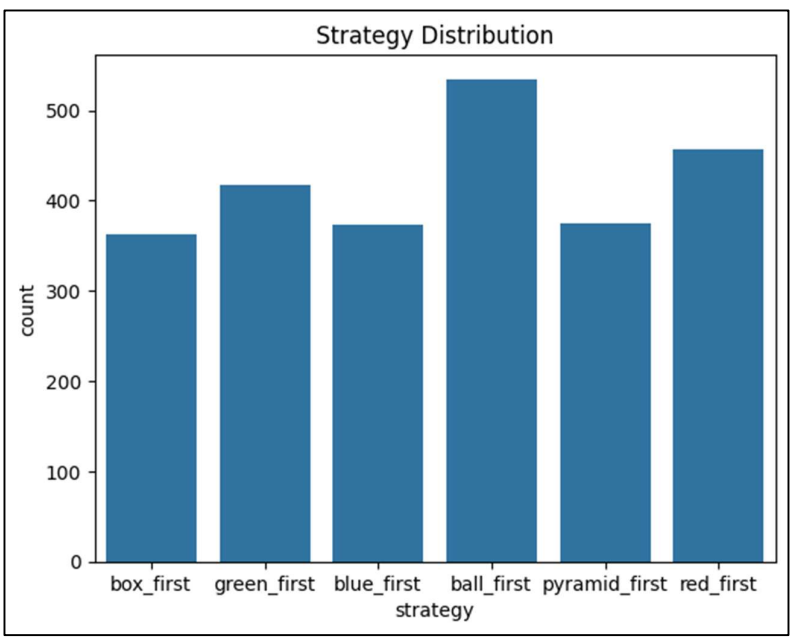
3.4.11 Feature Correlation Analysis

A correlation heatmap was used to analyze correlations between the six features. The results indicated a lack of significant correlations among the features, which indicated a lack of redundancy. Every feature offered unique information. This was another justification why an ensemble machine learning algorithm was appropriate.

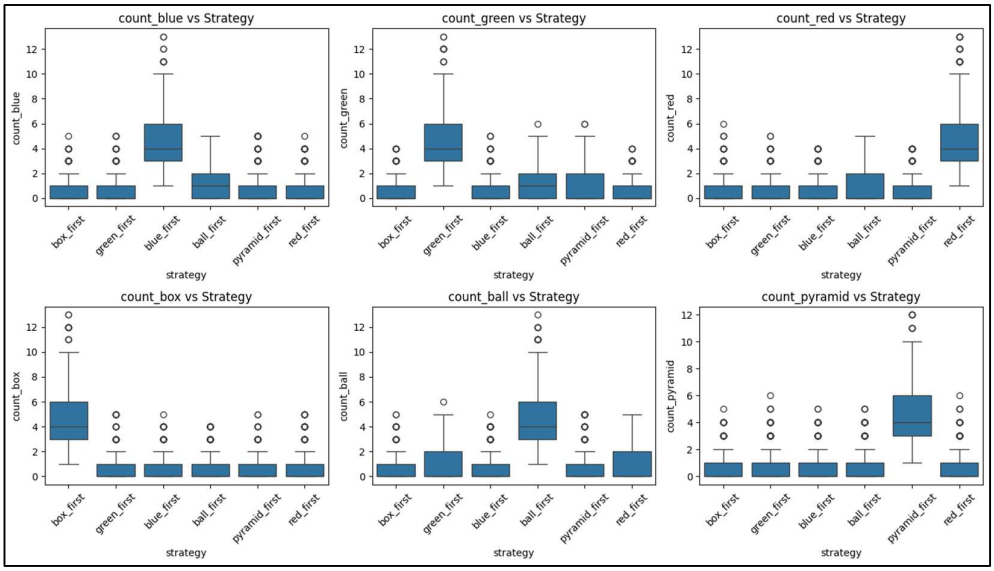
The figure illustrates the distribution of object count features across different sorting strategies, highlighting how variations in object composition influence strategy selection.

1. Sorting action Distribution

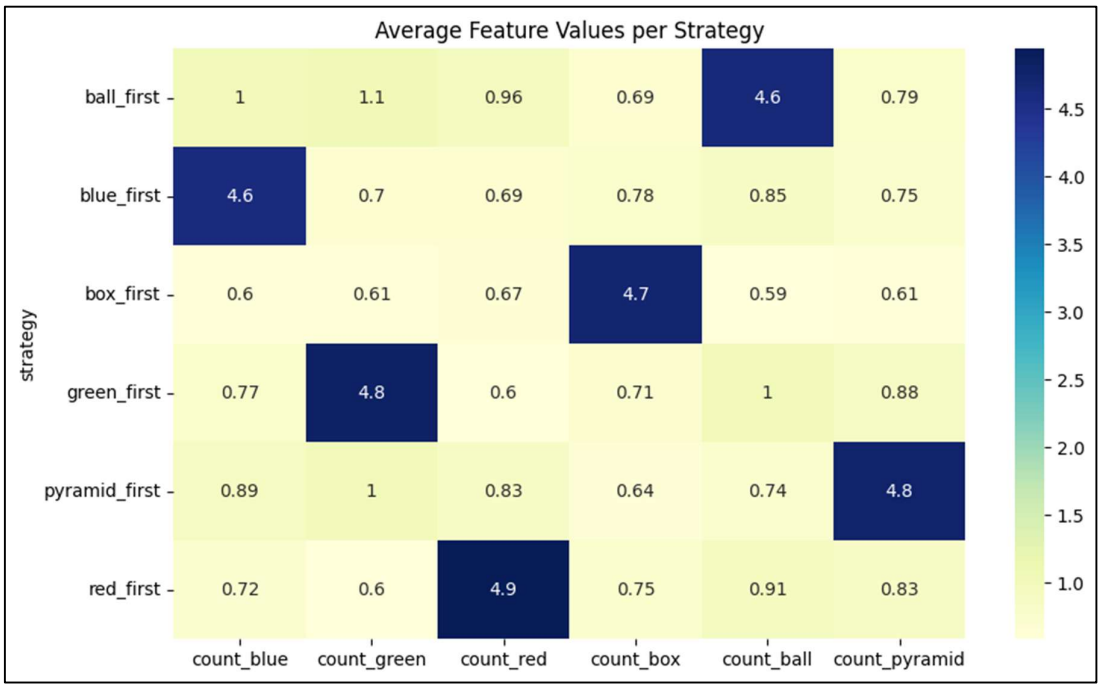
   The most used strategy was selecting the ball first.



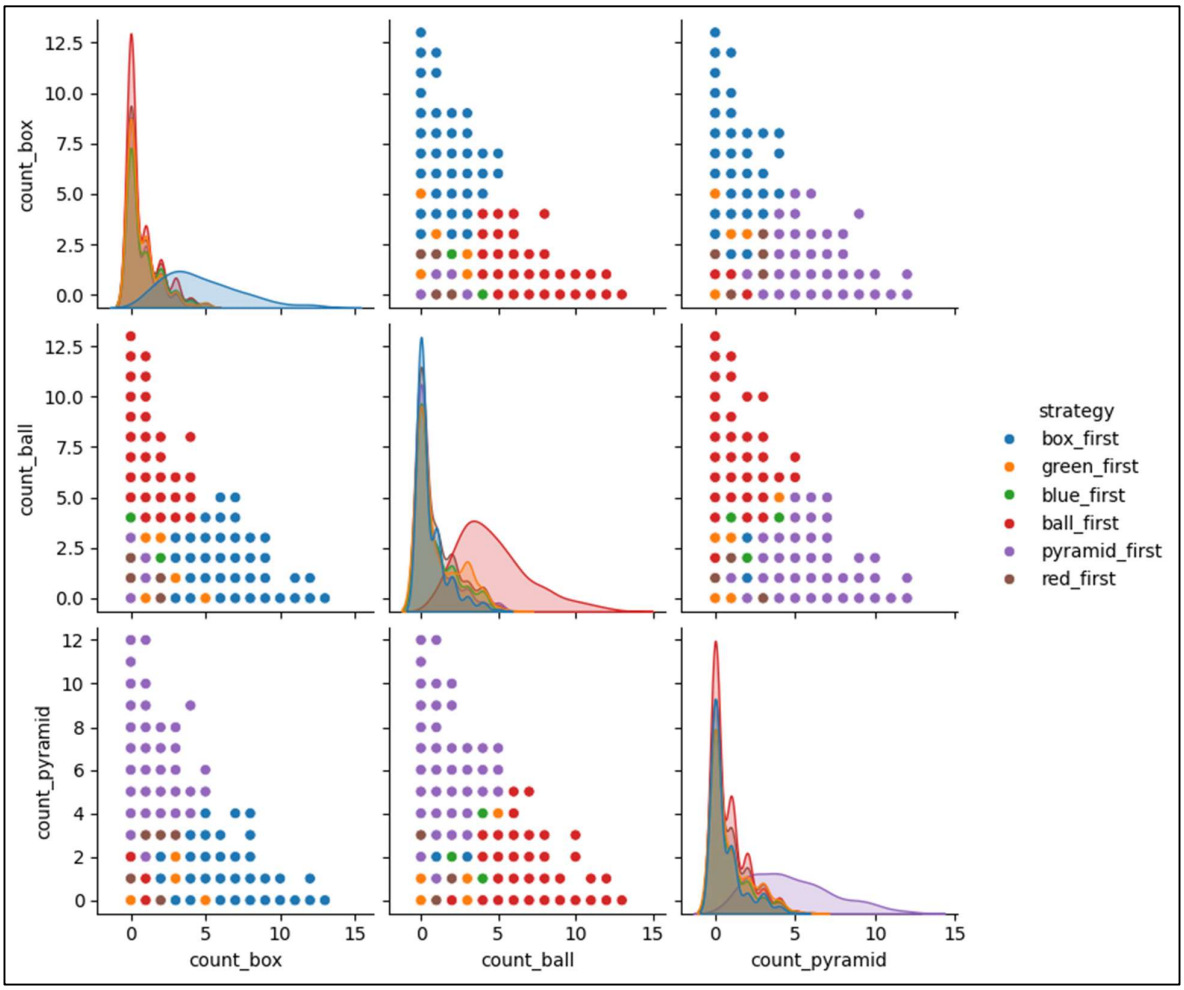
The figure illustrates the distribution of object count features across different sorting strategies, highlighting how variations in object composition influence strategy selection.



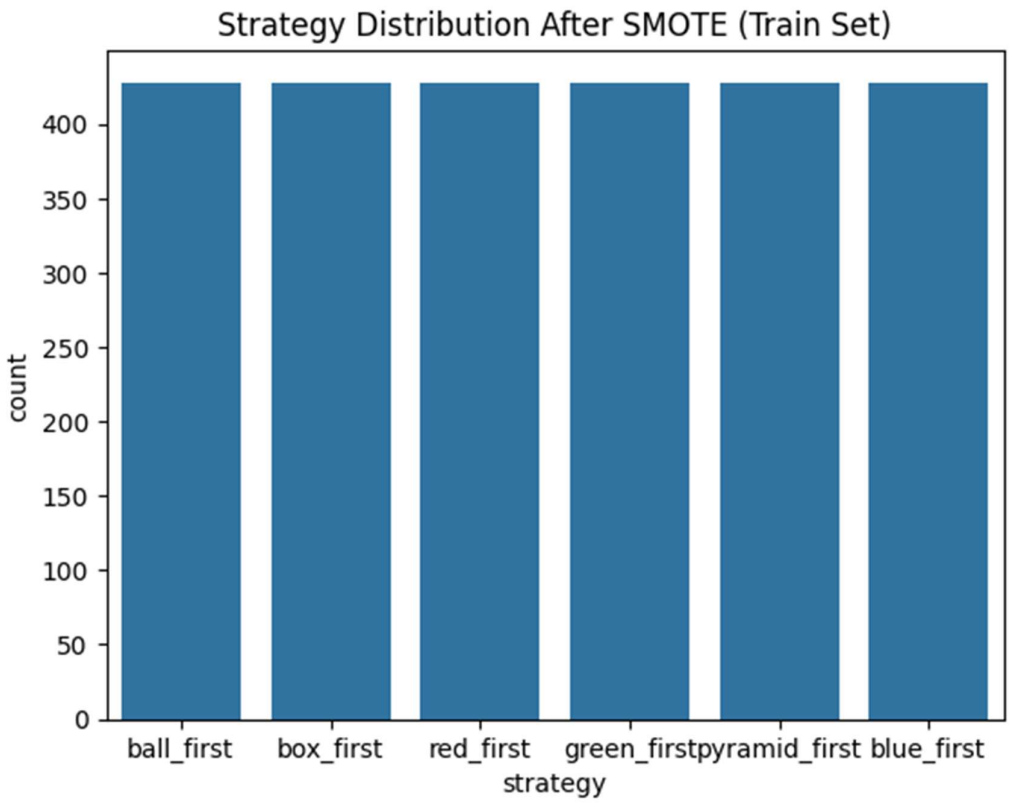
The heatmap reveals that strategies are predominantly selected when the average count of a corresponding object type exceeds approximately 4.5, indicating the influence of object dominance on strategy selection.



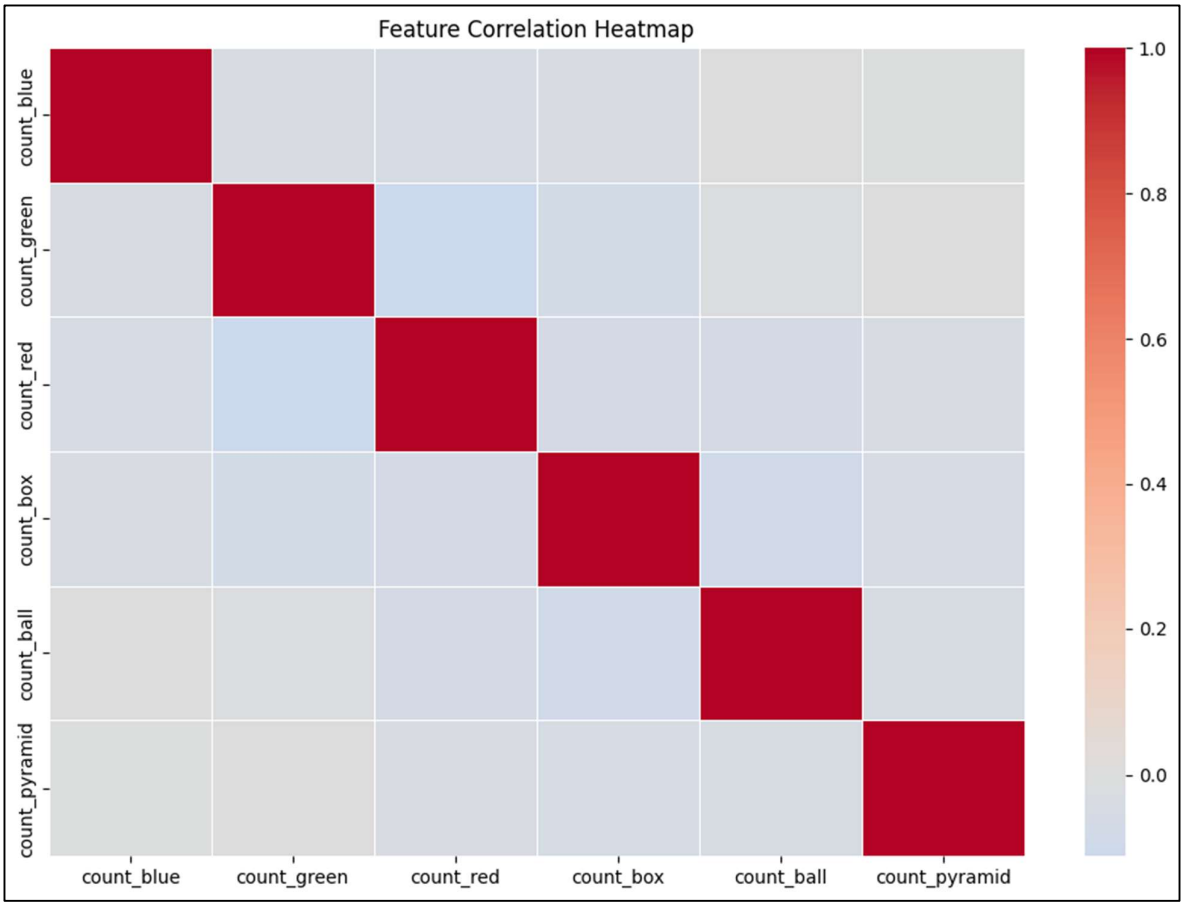
The pairplot illustrates the pairwise relationships between object shape counts, revealing strategy-specific clustering and overlapping regions in the feature space.



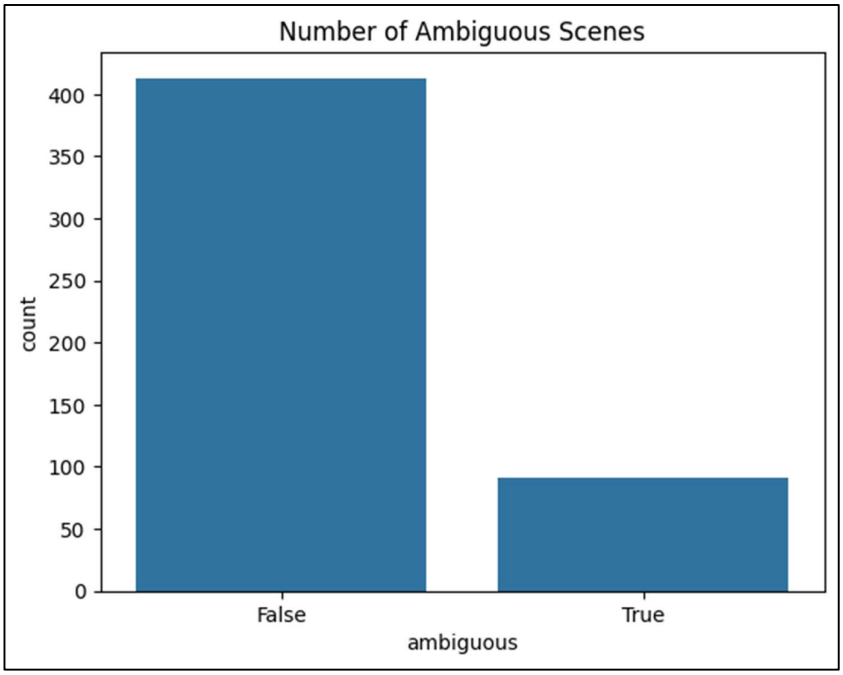
SMOTE was applied to address class imbalance, resulting in an approximately uniform distribution of strategies in the training data.



The feature correlation heatmap shows that color and shape counts are largely independent, with minimal correlation between features. This low redundancy ensures each feature contributes unique information, supporting stable model predictions and reliable SHAP explanations.



This plot shows the number of ambiguous scenes where no dominant object type is present, indicating inherently uncertain sorting scenarios. There are nearly 100 situations.



Model Selection and Training

The various machine learning models tested for strategy prediction include Logistic Regression, XGBoost, and Random Forest classifiers.

The Random Forest model was chosen based on its ability to model non-linear decision boundaries, robustness to noisy count-based features, avoidance of overfitting, and Intrinsic interpretability via feature importance measures.

The selected model was trained using the SMOTE-balanced training dataset to achieve improved predictive performance.

Model Evaluation Methodology

Evaluation based on confusion matrix Model performance was analyzed by making use of a confusion matrix, showing the accuracy of the prediction and pattern of misclassification for each strategy. It gives detailed insight into how the model performs across different classes of strategy.

Error Analysis with Ambiguity Filtering

An ambiguity filtering mechanism was introduced to distinguish genuine model errors from inherently ambiguous scenes. Scenes with nearly equal object dominance were excluded from the error analysis, as such cases are at best very difficult even for humans to classify.

The rest of the misclassified samples are already true model limitations. Taking a closer look at these cases gave them a better understanding of the conditions under which the model fails despite clear dominance patterns.

Feature-Based Error Inspection

This was further investigated with feature-level boxplots for incorrect predictions alone. Indeed, the boxplots showed that misclassifications occurred most often in scenes without any one feature strongly dominating others.

In addition, shape-based and color-based dominance for those misclassified samples were compared to determine whether the uncertainty arises more from shape attributes or from color attributes. These inspections helped in finding sources of ambiguity in feature space.

Model Persistence and Deployment Preparation

The trained Random Forest model as well as its feature list were serialized for storing consistency between the two environments. Using serialize, it is now easier for me to reuse the model for real-time inferencing.

Real-Time Strategy Recommendation and XAI Integration

A live video feed is obtained through the use of the ESP32 camera. Every frame passing through the feed is analyzed using the YOLOv8 perception algorithm to detect objects. Scene-level features are then obtained after which the suggested sorting method is determined using the trained Random Forest modeling.The implemented locking mechanism of a strategy locks strategies from being repeatedly switched. The strategies will stay active in the system for as long as all the related objects of a particular strategy have been cleared from the scene.

The explainability of this model is accomplished by using a technique called SHapley Additive exPlanations (SHAP). The feature values of SHAP are calculated based on the strategy chosen. The result of this calculation gives explanations in a human-understandable form. The features of this model recognize the object that has the most count values contributing to.

4. Modelling and Results

4.1. Results

Collaborative Texture Recognition

Table 4. 1 Results

Model	Accuracy
Sequential CNN	0.44
EfficientNet	0.88
MobileNet	0.89
DINOv2	0.98

- DINOv2 significantly outperforms CNN models
- CNNs struggle with similar textures
- Ultrasonic sensor improves reliability
- System works in real-time with low latency

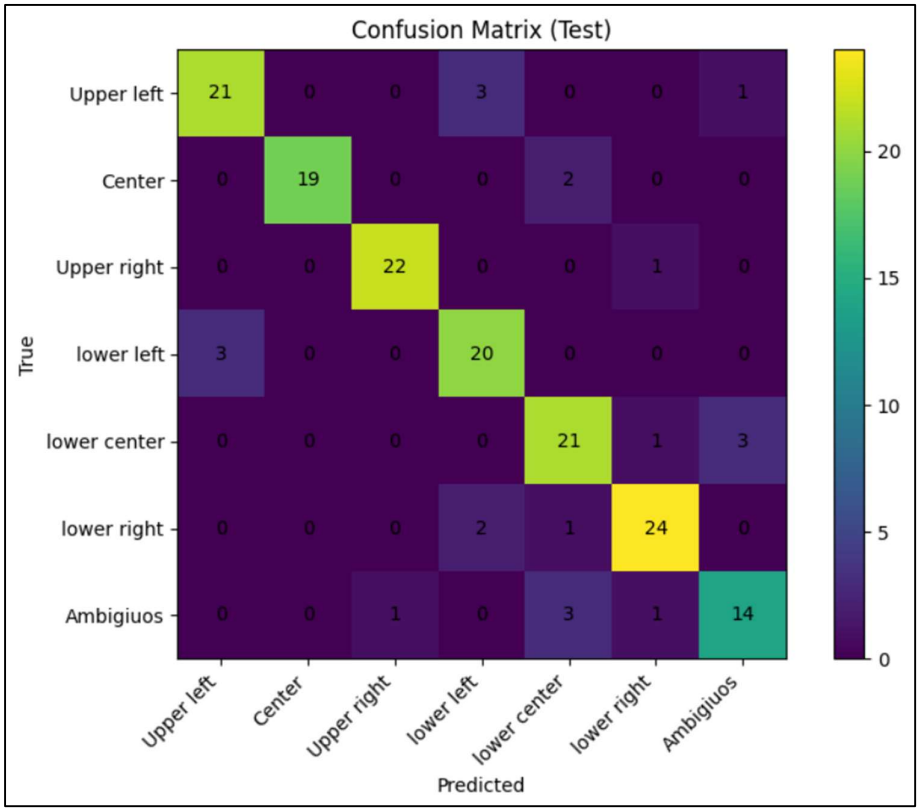
Adaptive Pattern Recognition via Human Behavior

The YOLO-based object detection model achieved strong performance, with a precision of 87%, recall of 81.7%, and a mean average precision (mAP@0.50) of 90%. These results indicate that the model is highly effective in detecting objects in real-time environments, although performance slightly decreases under challenging conditions such as poor lighting or occlusion.

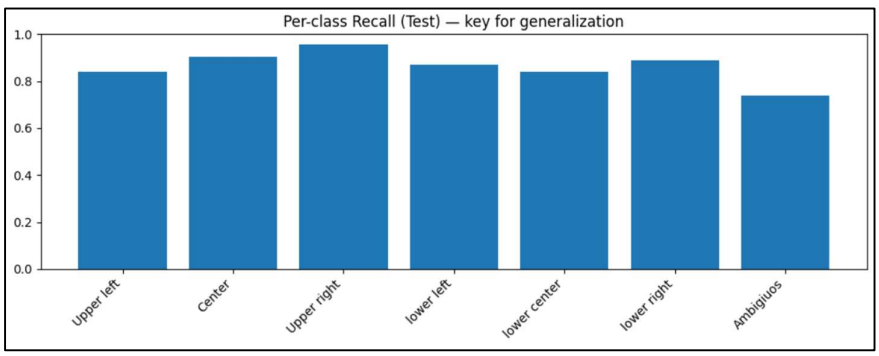
For speech classification, multiple models were evaluated, including Logistic Regression, Naive Bayes, and Decision Tree. Logistic Regression achieved the highest accuracy of 96%, making it the most suitable model for this task. Its performance is attributed to its ability to handle structured token-based features and linearly separable data effectively.

The integrated system demonstrated stable real-time performance, successfully combining object detection, speech recognition, and robotic interaction. The multimodal approach significantly improved user engagement and learning effectiveness by providing immediate feedback through visual, verbal, and physical channels.

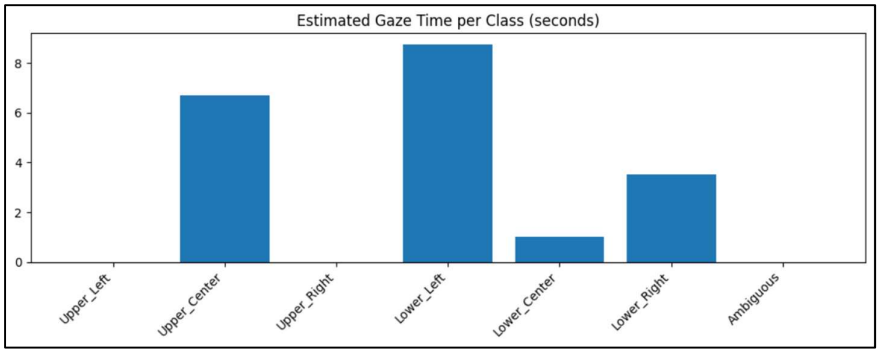
Human-Led Selection Through Gaze-Guided Robotic Assistance

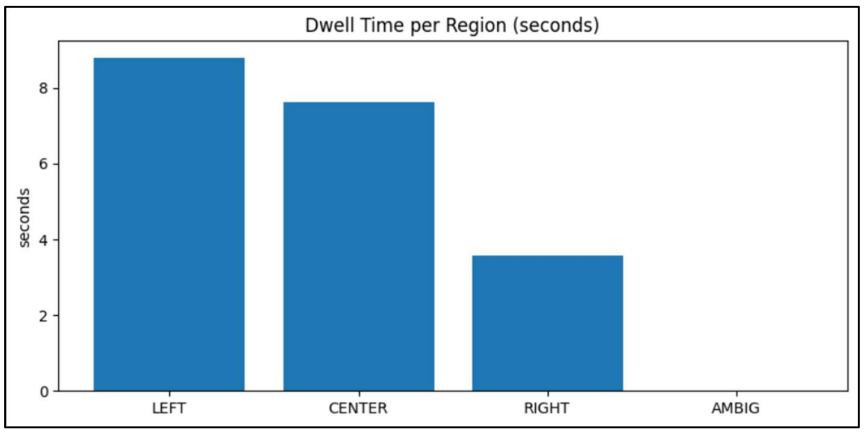


```
Classification report:
              precision    recall  f1-score   support

   Upper left     0.8750    0.8400    0.8571        25
       Center     1.0000    0.9048    0.9500        21
  Upper right     0.9565    0.9565    0.9565        23
   lower left     0.8000    0.8696    0.8333        23
 lower center     0.7778    0.8400    0.8077        25
  lower right     0.8889    0.8889    0.8889        27
    Ambigiuos     0.7778    0.7368    0.7568        19

     accuracy                         0.8650       163
    macro avg     0.8680    0.8624    0.8643       163
 weighted avg     0.8681    0.8650    0.8657       163
```



Inference on new images





Integrating Explainable AI and Adaptive Recommendations for Human Centered Robotic Sorting

The system was evaluated across spatial mapping, object detection, machine learning decision-making, and robotic execution. The homography-based grid mapping successfully transformed camera coordinates into grid coordinates, enabling accurate identification of object positions, mapping to robotic pick points, and maintaining spatial consistency across varying camera perspectives. This ensured that each detected object could be both classified and localized within the grid for further processing.

The performance of the object detection component was evaluated using the YOLOv8 model trained for 50 epochs. The results are presented in Table 4.2

Table 4. 2. Results of Yolo

Metric	Value
Precision	0.949
Recall	0.956
mAP@50	0.954
mAP@50–95	0.543

The model achieved a precision of 0.949 and recall of 0.956, indicating high accuracy with minimal false positives and false negatives. The mAP@50 value of 0.954 confirms strong detection performance, while the lower mAP@50–95 value of 0.543 indicates reduced accuracy under stricter localization thresholds. Overall, the model demonstrated reliable object detection suitable for real-time robotic applications.

For the recommendation system, three machine learning models were evaluated to determine the optimal sorting strategy. Their performance is summarized in Table 4.3

Table 4. 3

Model	Accuracy
Random Forest	0.94
XGBoost	0.88
Logistic Regression	0.88

Among the tested models, Random Forest achieved the highest accuracy of 0.94, outperforming both XGBoost and Logistic Regression, which showed similar performance. Based on these results, Random Forest was selected as the final model. A detailed evaluation of the Random Forest model is shown in Table 4.4

Table 4. 4

Metric	Value
Accuracy	0.94
Precision	0.93
Recall	0.74
F1-score	0.82

The Random Forest model demonstrated high precision (0.93), indicating reliable predictions, while the recall (0.74) suggests some limitations in identifying less frequent classes. The F1-score of 0.82 confirms a good balance between precision and recall, making it suitable for the recommendation task. In addition to accuracy, the model was selected due to its stability and compatibility with Explainable AI techniques, allowing better interpretation of decision-making.

4.2.Challenges Faced During Data Analysis

One of the major difficulties encountered during data analysis in this research was related to the multimodal and heterogenous character of the system used. Specifically, in this project it was required to combine data obtained from multiple modes including color images, frames from a closed-circuit television, speech signals, readings from ultrasonic sensors, and behavioral interaction data in a single integrated human-robot collaboration. As these types of data have differences in their representation and sampling rate, as well as in their level of complexity, it was not easy to synchronize and process this data effectively.

Another significant problem encountered in this study was connected with a high degree of variability observed in visual datasets collected. This included images recorded under different light conditions, using various angles and distances between a person and the object, as well as under diverse background and even differences between an RGB image and CCTV frame. This variety made models generalization difficult as well as performed tasks such as material recognition, object detection, and gaze estimation. Material classification became especially problematic as different objects including plastic, glass, metal, and rubber have many similar visual features including texture or reflectance.

Imbalanced data and poor representation of some conditions were among the other challenges faced in the analysis process. Some object pairings, gaze orientations, and speech reactions were poorly represented, affecting how the model learned from them and necessitating the application of data augmentation and balancing methods like SMOTE. Furthermore, the human-made data was inevitably ambiguous, particularly in the case of gaze-based region selection and sorting method classification, due to unclear boundaries in ground truths.

The real-time nature of the system presented additional challenges, given that several models needed to run concurrently without high latency to facilitate interaction. It was necessary to optimize the models, choosing light models for better efficiency. Other sources of noise in the data included blurring of images, background noise in the audio data, and inconsistencies in the sensor data.

Fourthly, there were issues related to feature overlapping and uncertainty during decision-making, especially when dealing with the strategy recommendation module. In most cases, several object features exhibited comparable dominance scores, making rule-based methods less effective and necessitating the use of advanced machine learning algorithms. Moreover, ensuring model interpretability was another difficulty since incorporating SHAP explanations increased computational costs. The primary challenge, however, was the seamless incorporation of varied, complex, and sometimes ambiguous data sets into an efficient human-robot collaboration framework.

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